

KRISH GUPTA

+91-9157557166

guptakrish490@gmail.com

linkedin.com/in/krish-gupta

github.com/guptakrish490

EDUCATION

Indian Institute of Information Technology Sri City (IIIT Sri City)

August 2025 – August 2029

Bachelor of Technology in Computer Science and Engineering

CGPA: 8.35/10

Relevant Coursework: Data Structures, Algorithms, Object-Oriented Programming, Operating Systems, Database Management Systems

TECHNICAL SKILLS

Languages: C++, C, TypeScript, JavaScript, SQL

Frontend: React.js, HTML5, CSS3, Tailwind CSS

Backend: Node.js, Fastify, Express.js, REST APIs

Database: PostgreSQL, MongoDB

Developer Tools: Git, GitHub, VS Code, Postman

PROJECTS

FlowForge – Fault-Tolerant Async Job Processing System

GitHub

TypeScript, Fastify, Node.js, PostgreSQL, Git

Sep 2026 – Oct 2026

- Built a concurrent asynchronous job processing system in TypeScript and PostgreSQL, with a Fastify API, worker processes, configurable concurrency, buffering, retries, and failure recovery.
- Designed atomic job claiming and ownership-aware updates using PostgreSQL transactions and row-level locking, preventing duplicate processing across concurrent workers.
- Implemented worker ownership tracking, lease-based stale-job recovery, retry handling, and ownership-aware status updates to prevent recovered jobs from being overwritten by stale workers.
- Tracked the job lifecycle across creation, claiming, execution, completion, and recovery using timestamps and worker metadata to measure queue wait time and processing latency.
- Benchmarked 5,000-job workload across worker, concurrency, and buffer configurations, cutting processing time from ~1,511.8s to ~12.04s (~3.3 → ~415 jobs/sec, a ~125x gain over the single-worker baseline).

DevTrack Pro – MERN-Based Developer Productivity Platform

GitHub | Live Demo

React.js, Node.js, Express.js, MongoDB, JWT, Zod, Tailwind CSS

Jun 2026 – Jul 2026

- Developed a modular full-stack productivity platform with 14+ REST APIs for managing goals, projects, tasks, developer activity, and user profiles.
- Implemented JWT-based authentication with refresh-token sessions, HTTP-only cookies, authorization middleware, Zod validation, rate limiting, protected routes, and bcrypt password hashing.
- Built automatic activity logging, streak tracking, search, pagination, and dashboard analytics across goals, projects, tasks, and recent activity.
- Optimized MongoDB read operations using lean queries and pagination, and deployed the application on Vercel and Render with MongoDB Atlas.

PROBLEM SOLVING

- Solved 200+ Data Structures and Algorithms problems on LeetCode.
- Strong fundamentals across core topics: Arrays, Linked Lists, Stacks, Queues, Trees, Graphs, Hashing, Binary Search, BFS/DFS, and Topological Sorting.

ACHIEVEMENTS

- IOTA Club Web Development Domain: Built and contributed to IOTA club's website and handled deployment.
- Open Source Contributor – freeCodeCamp: Merged a contribution to the Node.js curriculum, improving the documentation/explanation of `globalThis`.
- Qualified JEE Mains with AIR 28,858 out of more than 1.6 million participating candidates across the nation.
- Selected for the 1-year Dakshana JDST (JNVs) scholarship among 650 students in India, granting access to top-tier JEE coaching and elite mentorship.
- Earned the NCC 'A' Certificate, demonstrating discipline in structured environments, team leadership, and civic duty.